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for Physics of Nanostructures

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"Adaptive and Robust Control"

on the topic:
STATE FEEDBACK ADAPTIVE CONTROL

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INTRODUCTION

0.1 State Feedback Adaptive Control

Problem statement. Consider the plant given in the state-space form:

$$\dot{x} = Ax + bu, \quad x(0) = x_0, \quad (3.1)$$

$$y = c^T x, \quad (3.2)$$

where $x \in \mathbf{R}^n$ is the state vector, $u \in \mathbf{R}$ is the control input, and $y \in \mathbf{R}$ is the regulated output. The matrix $A \in \mathbf{R}^{n \times n}$ is an unknown constant matrix, while $b, c \in \mathbf{R}^n$ are known constant vectors.

The objective is to design a control law that compensates the parametric uncertainty and guarantees

$$\lim_{t \rightarrow \infty} (x_m(t) - x(t)) = 0, \quad \text{i.e.,} \quad \lim_{t \rightarrow \infty} e(t) = 0, \quad (3.3)$$

where $e = x_m - x$ is the vector of control errors, and $x_m \in \mathbf{R}^n$ is the state of the reference model

$$\dot{x}_m = A_M x_m + b_M g(t), \quad (3.4)$$

$$y_m = c_M^T x_m, \quad (3.5)$$

with $g(t)$ being a bounded piecewise-continuous reference signal, A_M a Hurwitz matrix, and b_M, c_M known constant vectors.

Assumption 3.1 (Matching conditions). There exist a vector $\theta \in \mathbf{R}^n$ and a scalar κ such that

$$A = A_M + b\theta^T, \quad b_M = \kappa b. \quad (3.6)$$

This assumption allows the use of the certainty equivalence principle for deriving an adaptation law. For simplicity, the plant is considered in canonical form:

$$A = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & 0 & 1 \\ -a_n & -a_{n-1} & \cdots & -a_2 & -a_1 \end{bmatrix}, \quad b = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ b_0 \end{bmatrix}, \quad c = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}.$$

The reference model (A_M, b_M, c_M) has the same canonical structure, with parameters a_{Mi} chosen via pole placement to achieve the desired transient properties:

$$\theta_i = \frac{a_{Mi-1} - a_{i-1}}{b_0}, \quad \kappa = \frac{b_{M0}}{b_0}. \quad (3.7)$$

Problem solution. Assuming the parameters of A are known, a control law ensuring condition (3.3) can be designed using the error dynamics:

$$\dot{e} = A_M e + b \left(\frac{1}{\kappa} \theta^T x + \frac{1}{\kappa} u - g \right). \quad (3.8)$$

Choosing

$$u = \frac{1}{b_0} \theta^T x + g, \quad (3.9)$$

yields $\dot{e} = A_M e$, providing exponential convergence of $e(t)$ to zero.

Since θ is unknown, it is replaced with its estimate $\hat{\theta}$ to form the adjustable control law

$$u = \frac{1}{b_0} \hat{\theta}^T x + g. \quad (3.10)$$

Substituting (3.10) into (3.8) gives

$$\dot{e} = A_M e + b x^T \tilde{\theta}, \quad (3.11)$$

where $\tilde{\theta} = \theta - \hat{\theta}$ is the parameter estimation error.

Define the Lyapunov function candidate

$$V = \frac{1}{2} e^T P e + \frac{1}{2\gamma} \tilde{\theta}^T \tilde{\theta},$$

where P satisfies the Lyapunov equation

$$A_M^T P + P A_M = -Q, \quad (3.12)$$

with $Q = Q^T > 0$. Its time derivative is

$$\dot{V} = -\frac{1}{2} e^T Q e + \tilde{\theta}^T (x^T P b + \frac{1}{\gamma} \dot{\hat{\theta}}).$$

Choosing the adaptation law

$$\dot{\hat{\theta}} = \gamma x P^T b, \quad \hat{\theta}(0) = 0, \quad (3.13)$$

ensures $\dot{V} \leq 0$, which guarantees boundedness and convergence.

Properties. Under Assumption 3.1 and bounded $g(t)$, the adaptive control (3.10)–(3.13) guarantees:

- boundedness of all closed-loop signals;
- convergence $e(t) \rightarrow 0$ as $t \rightarrow \infty$;
- exponential convergence of $\tilde{\theta}(t)$ to zero if $x(t)$ satisfies the persistent excitation (PE) condition

$$\int_t^{t+T} x(\tau)x^T(\tau) d\tau \geq \alpha I, \quad (3.14)$$

for some constants $\alpha > 0$, $T > 0$;

- existence of an optimal adaptation gain γ that maximizes the convergence rate.

0.2 The order of the work

1. Using the desired transient time T_t and maximum overshoot σ from Table 2, design the reference model (3.4)–(3.5) via pole placement. If $\sigma = 0$, use a second-order binomial or Butterworth polynomial:

$$(s^2 + 2\omega_0 s + \omega_0^2) = 0,$$

where ω_0 is calculated from normalized transient time T'_t .

Plot the step response of the reference model and mark T_t and σ .

2. Assuming the plant parameters are known, simulate the closed-loop system with nonadaptive control (3.9) using parameters from (3.7). Perform three experiments:
 - nominal parameters (a_0, a_1) ,
 - slightly perturbed stable parameters,
 - destabilized parameters.

Plot $x(t)$ and $x_m(t)$, as well as the error $e(t)$.

3. Repeat the above experiments with the adaptive controller (3.10)–(3.13). Demonstrate the properties (boundedness, convergence, PE) through simulation. Perform trials with three different γ values, then repeat for $g(t) = 1$. Plot x , x_m , e , and $\tilde{\theta}$ for analysis.
4. Draw conclusions after each experiment.

0.3 Notations

Throughout this work, the following notations are used:

Table 1 — Table 1: Notations of symbols

Symbol	Explanation
A	Plant model
b_0	Coefficient
t_T	Transient time
e	Control error
x	Output of the plant
x_m	Output of the reference model
$\tilde{\theta}$	Parameter estimation error
g	Reference signal
λ	Parameter for transient performance
$\bar{\sigma}$	Maximum overshooting (%)

0.4 Variant

Table 2 — System parameters and reference signal

No.	Matrix A	Coeff. b_0	Transient time t_T	Maximum overshoot $\bar{\sigma}$, %	Reference $g(t)$
15	$\begin{bmatrix} 0 & 1 \\ -15 & -7 \end{bmatrix}$	3	0.7	0	$\text{sign}(\sin(0.5t)) + 2$

1 SOLUTIONS

1.1 Task 1 : Design the Reference Model

Consider the plant described by

$$\begin{cases} \dot{x} = Ax + b_0 u, \\ y = c^T x, \end{cases} \quad A = \begin{bmatrix} 0 & 1 \\ -15 & -7 \end{bmatrix}, \quad b_0 = 3, \quad c = \begin{bmatrix} 1 \\ 0 \end{bmatrix}. \quad (1)$$

The goal of this step is to design the reference model that determines the desired transient response of the adaptive control system according to the given variant data.

The reference model is defined as

$$\begin{cases} \dot{x}_M = A_M x_M + b_M g(t), \\ y_M = c_M^T x_M, \end{cases} \quad (2)$$

where $x_M \in \mathbf{R}^2$ is the model state vector, and $g(t)$ is the reference signal:

$$g(t) = \text{sign}(\sin(0.5t)) + 2.$$

Matrices of the reference model are given as

$$A_M = \begin{bmatrix} 0 & 1 \\ -a_{M0} & -a_{M1} \end{bmatrix}, \quad b_M = \begin{bmatrix} 0 \\ a_{M0} \end{bmatrix}, \quad c_M = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Pole–placement design

To provide the transient time $t_T = 0.7$ s and maximum overshoot $\bar{\sigma} = 0$, the desired characteristic polynomial of the reference model is chosen as the *binomial* second–order polynomial:

$$s^2 + 2\omega_0 s + \omega_0^2 = 0. \quad (3)$$

For zero overshoot, the damping ratio is $\xi = 1$. Using the normalized relationship between transient time and natural frequency,

$$\omega_0 = \frac{4.8}{t_T}, \quad (4)$$

we obtain

$$\omega_0 = \frac{4.8}{0.7} = 6.86.$$

Therefore, the coefficients of the reference model are

$$a_{M1} = 2\omega_0 = 13.72, \quad a_{M0} = \omega_0^2 = 47.07. \quad (5)$$

Hence, the reference model matrices are explicitly:

$$A_M = \begin{bmatrix} 0 & 1 \\ -47.07 & -13.72 \end{bmatrix}, \quad b_M = \begin{bmatrix} 0 \\ 47.07 \end{bmatrix}, \quad c_M = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

The corresponding transfer function from $g(t)$ to $y_M(t)$ is:

$$h(t) = \frac{47.07}{s^2 + 13.72s + 47.07}. \quad (6)$$

This reference model provides a fast critically-damped response without overshoot and defines the desired transient performance for the adaptive control design.

1.1.1 Simulink Model

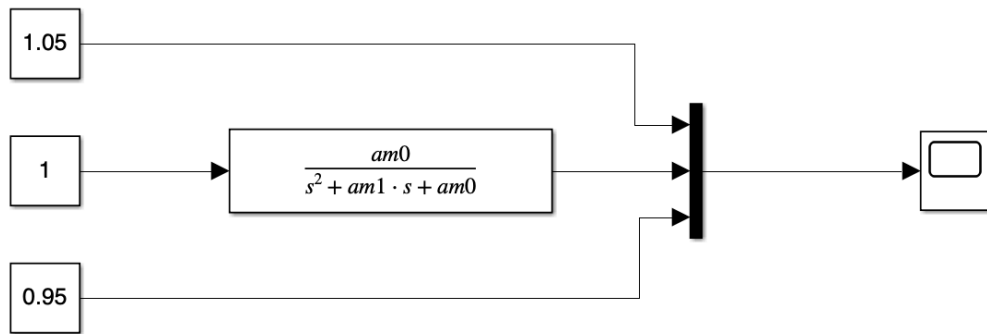
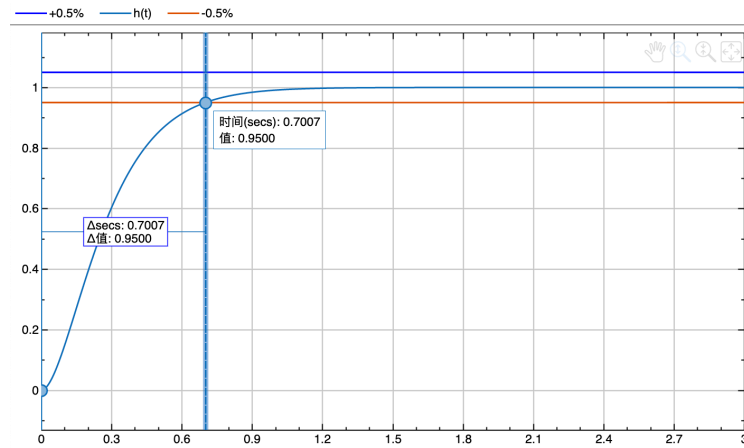


Figure 1 — Simulink implementation of the reference model (2).



(a) Step response of the designed reference model showing $t_T = 0.7$ s and $\bar{\sigma} = 0$.

1.1.2 Conclusion

The reference model was successfully designed based on the parameters of Variant 15. It exhibits a critically damped response with no overshoot and a transient time of approximately 0.7 s. This model will serve as the desired trajectory for adaptive and non-adaptive control simulations in subsequent tasks.

1.2 Task 2 : Design and simulate the closed-loop system with non-adaptive control

In this stage, the control law is designed under the assumption that all plant parameters are known. From the plant model

$$\dot{x} = Ax + bu,$$

and the matching condition

$$A = A_M + b\theta^T, \quad b_M = \kappa b,$$

the non-adaptive (certainty-equivalent) control law can be written as:

$$u = \frac{1}{b_0}\theta^T x + g(t). \quad (7)$$

Substituting (7) into the error dynamics gives:

$$\dot{e} = A_M e,$$

which ensures exponential convergence $e(t) \rightarrow 0$.

1.2.1 Simulink Model

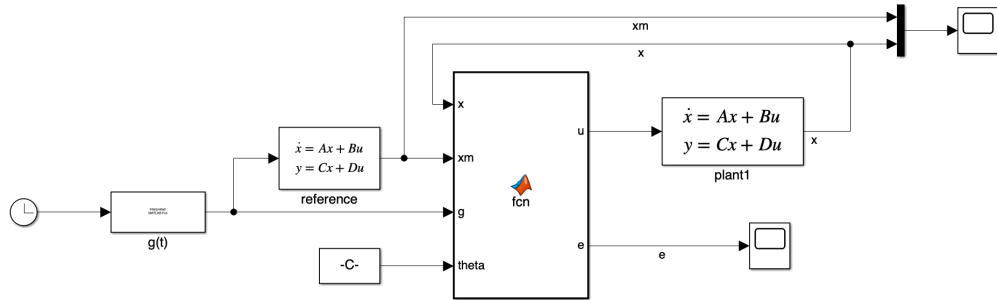


Figure 3 — Simulink implementation of the non-adaptive control system.

1.2.2 Conclusion

The non-adaptive controller achieves accurate tracking as long as the plant parameters remain fixed and correctly identified. However, when parameter uncertainties or external disturbances appear, the system performance may degrade, motivating the development of adaptive control in Task 3.

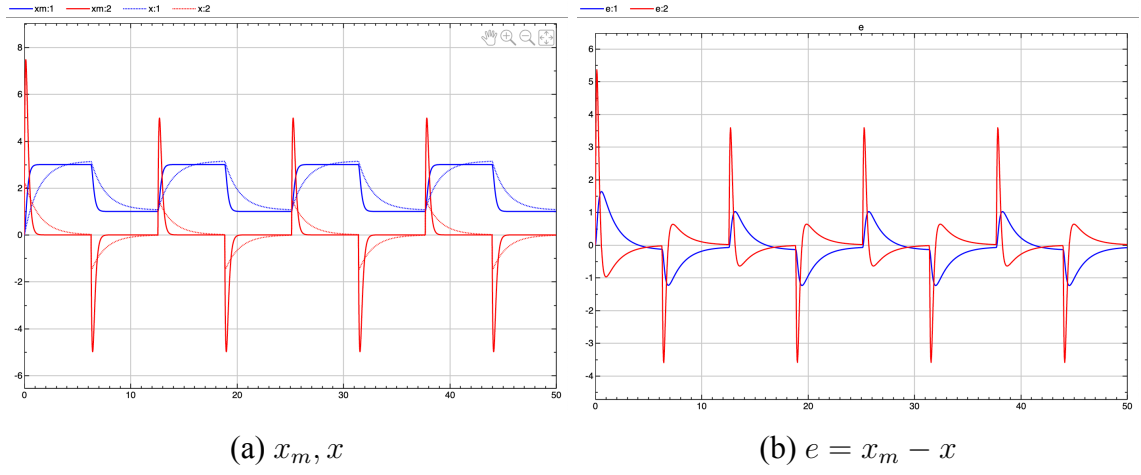


Figure 4 — System performance under known parameters.

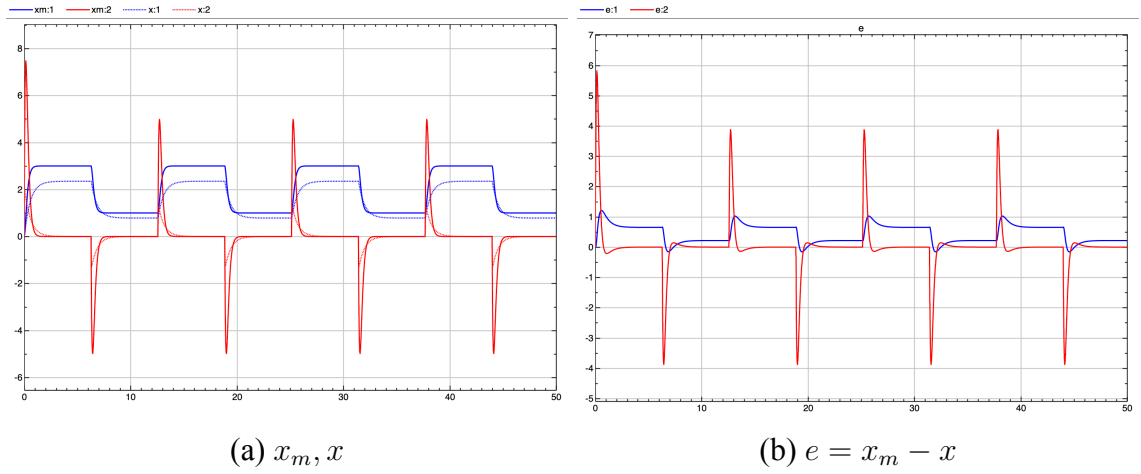


Figure 5 — $A = 2 * A$

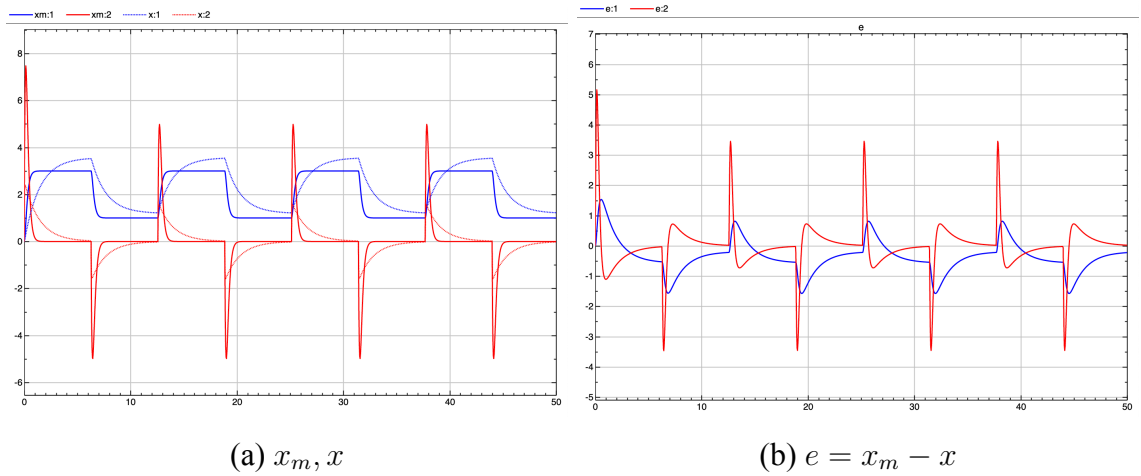


Figure 6 — $A = A + \begin{bmatrix} 0 & 0 \\ 5 & 5 \end{bmatrix}$

1.3 Task 3 : Repeat the experiments

1.3.1 Repeat this triple of experiments with adaptive controller

In this part, the adaptive control law and parameter update rule are introduced to handle the case when θ is unknown:

$$u = \frac{1}{b_0} \hat{\theta}^T x + g(t), \quad (8)$$

$$\dot{\hat{\theta}} = \gamma x P^T b, \quad (9)$$

where P is the positive definite solution of the Lyapunov equation

$$A_M^T P + P A_M = -Q.$$

The adaptation gain $\gamma > 0$ determines the rate of parameter convergence. The closed-loop stability is proven using the Lyapunov function:

$$V = \frac{1}{2} e^T P e + \frac{1}{2\gamma} \tilde{\theta}^T \tilde{\theta}, \quad \dot{V} = -\frac{1}{2} e^T Q e \leq 0.$$

1.3.2 Simulink Model

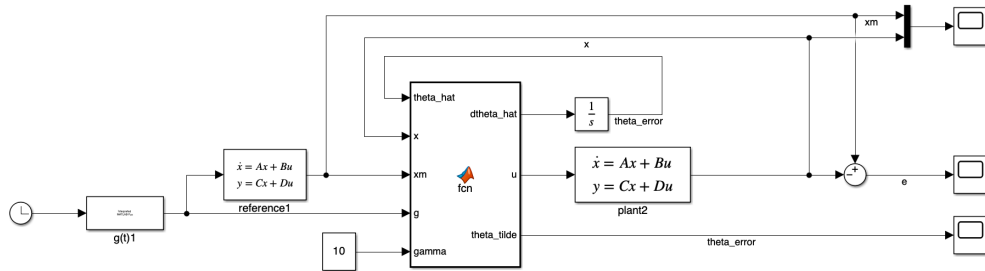
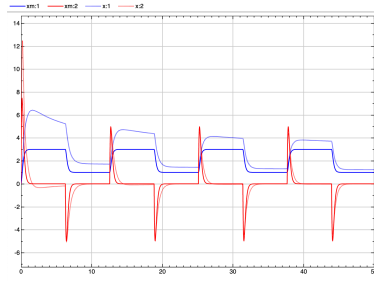
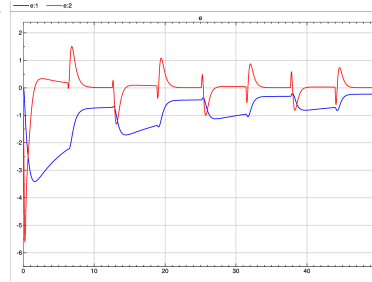
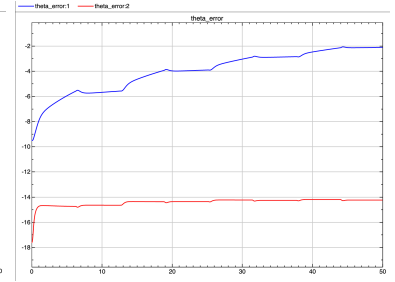
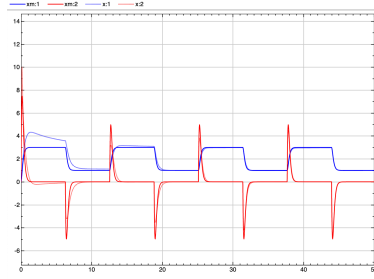
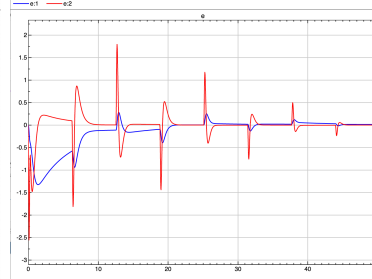
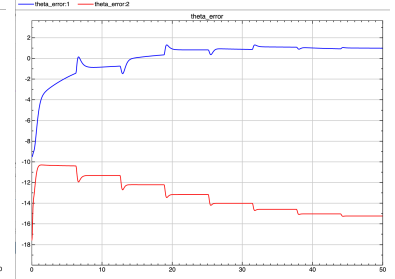
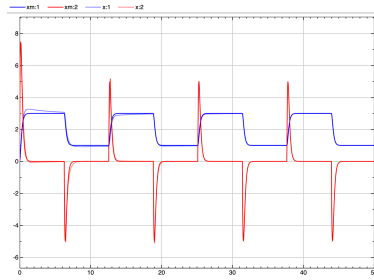
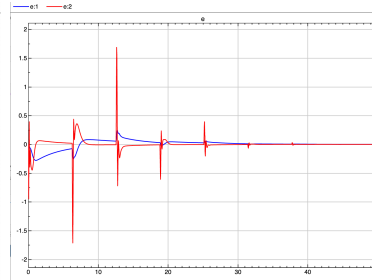
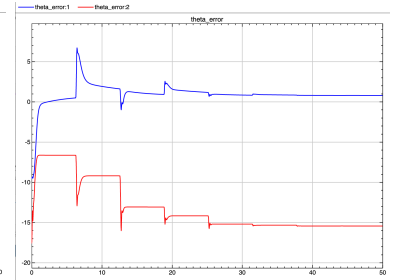


Figure 7 — Simulink model of the adaptive control system implementing Eqs. (8)–(9).

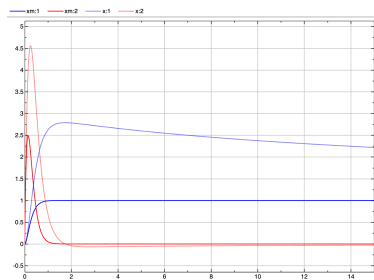
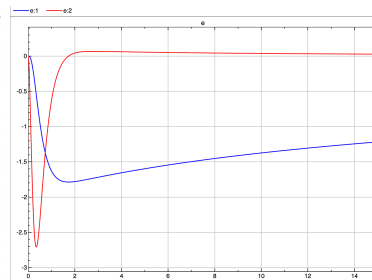
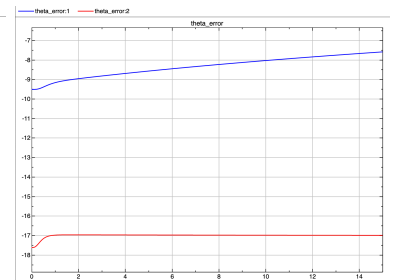
1.3.3 Effect of adaptation gain γ

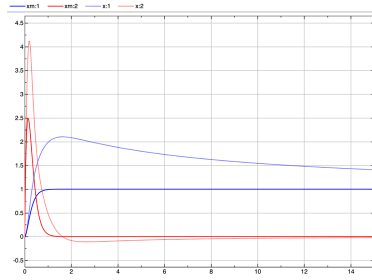
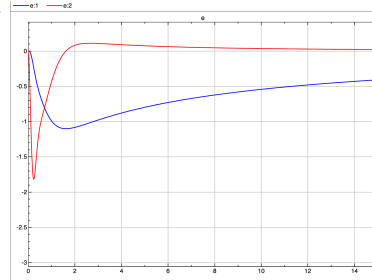
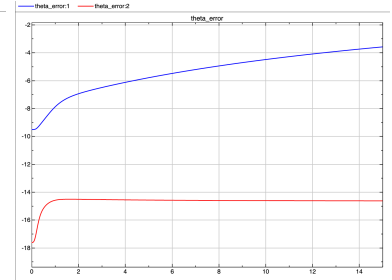
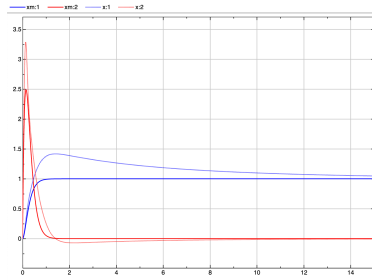
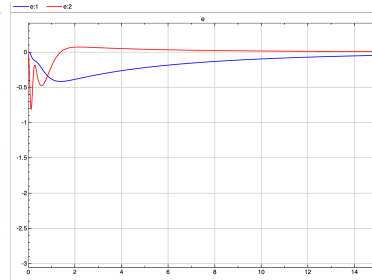
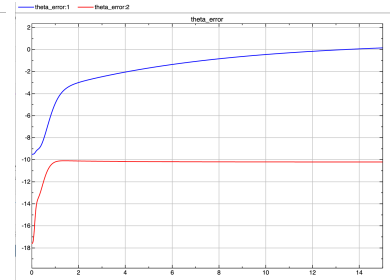
Repeat simulations for three values of γ (e.g., 1, 10, 100). A larger γ accelerates convergence of $\hat{\theta}$ but may induce oscillations in $e(t)$ due to aggressive adaptation. The trade-off between convergence speed and smoothness should be highlighted.


(a) $x_m - x, \gamma = 1$

(b) $e = x_m - x, \gamma = 1$

(c) $\tilde{\theta}(t), \gamma = 1$

(a) $x_m - x, \gamma = 10$

(b) $e = x_m - x, \gamma = 10$

(c) $\tilde{\theta}(t), \gamma = 10$

(a) $x_m - x, \gamma = 100$

(b) $e = x_m - x, \gamma = 100$

(c) $\tilde{\theta}(t), \gamma = 100$

1.3.4 Repeat the experiment for $g(t) = 1$

The constant reference $g(t) = 1$ verifies the steady-state tracking ability of the adaptive controller. Under persistent excitation conditions, both $e(t) \rightarrow 0$ and $\tilde{\theta}(t) \rightarrow 0$.


(a) $x_m - x, \gamma = 1$

(b) $e = x_m - x, \gamma = 1$

(c) $\tilde{\theta}(t), \gamma = 1$


(a) $x_m - x, \gamma = 10$

(b) $e = x_m - x, \gamma = 10$

(c) $\tilde{\theta}(t), \gamma = 10$

(a) $x_m - x, \gamma = 100$

(b) $e = x_m - x, \gamma = 100$

(c) $\tilde{\theta}(t), \gamma = 100$

1.3.5 Conclusion

The adaptive control system effectively compensates for parameter uncertainty and guarantees asymptotic tracking of the reference model. Simulation results show that an appropriate choice of γ achieves fast adaptation without oscillations, confirming the theoretical properties derived from the Lyapunov analysis.